MISSION-READY | FULL-STACK INTEGRATED SYSTEM | NDMA PILOT

ReliefRoute: Autonomous Emergency Logistics System

Closing the 90-Minute Dispatch Gap: From Raw SOS Signals to Hazard-Bypassing Last-Mile Relief in Under 120 Seconds

OPERATIONAL CRISIS

- Control room paralysis during extreme urban flash floods.
- Disjointed telephone trees, paper logs, and manual spreadsheets.
- 90–180 min delays breach the life-critical 'Golden Hour' of rescue.

AUTONOMOUS CORE

- Real-time multilingual SOS signal ingestion (audio/text/web).
- Sub-second LangGraph multi-agent logistical entity extraction.
- Algorithmic depot matching with dynamic inventory reservation.

ENTERPRISE INTEGRATION

- Next.js 16 dark-mode tactical cockpit with MapLibre GL spatial UI.
- FastAPI asynchronous microservices orchestrating Python state machines.
- ACID-backed Neon PostgreSQL persistence with Telegram Bot gateway.

THE STRATEGIC BOTTLENECK

The Lethal 90-Minute Dispatch Paralysis in Urban Floods

Information Deluge, Blind Navigation, and Manual Bottlenecks Break the Golden Hour of Rescue

1. THE SOS INGESTION DELUGE

- Flash floods trigger 5,000+ frantic citizen messages across Telegram & WhatsApp.
- Raw messages lack structured GPS coordinates and standardized commodity requests.
- Exhausted human dispatchers drown in noisy, unverified natural language text.

2. THE BLIND NAVIGATION TRAP

- Commercial GPS (Google Maps, standard fleet apps) has zero floodwater awareness.
- Aid trucks and ambulances are dispatched directly into submerged arterial roads.
- Results in drowned rescue vehicles, lost aid cargo, and stranded emergency teams.

3. SILOED LOGISTICS FRICTION

- Red Cross, state food stockpiles & pharma warehouses operate on disconnected DBs.
- Aid is dispatched blindly: dry rations sent where infants need electrolytes or insulin.
- Manual assignment consumes 90–180 minutes per truck roll-out.

THE ARCHITECTURAL SOLUTION

Gen-AI Command Console: Raw SOS to Actionable Dispatch

Converting Panicked Citizen Distress Streams into Structured, Validated Logistical Plans in Seconds

MULTILINGUAL INGESTION

- Ingests unstructured citizen voice notes and texts across Telegram and Web.
- Handles emotional panic, dialectal expressions, and fragmented descriptions.
- Instant conversational auto-prompting for missing landmark details.

SUB-SECOND EXTRACTION

- LangGraph multi-agent extraction pipeline outputs structured manifests in <1.2s.
- Extracts verified coordinates, affected headcount, and prioritized supplies.
- Calculates 98%+ extraction confidence metrics for operator review.

ALGORITHMIC RESERVATION

- Directly queries active regional depots and pharmaceutical hubs.
- Programmatically locks exact stock (e.g., 500L water, 120 ORS, rafts) with zero lag.
- Synthesizes complete, executable DeliveryPlan records with turn-by-turn vectors.

SAFETY & GOVERNANCE

Hazard-Aware Routing & Human-in-the-Loop Safety Locks

Dynamic Hydrological Ingress Avoidance Paired with Deterministic Command Safeguards

HYDROLOGICAL HAZARDS

- Ingests live Open-Meteo precipitation models and real-time sensor radar.
- Generates dynamic GeoJSON polygonal flood barriers across river & lake basins.
- Routing engine treats flood contours as impassable physical obstacles.

AUTONOMOUS DETOURS

- OpenRouteService spatial integration dynamically plots curved safety perimeters.
- Live demo PLAN-2024-003: critical insulin delivery detour arcs 2.2km around basin.
- Guarantees safe passage clearance buffers away from rapid flood ingress.

PENDING APPROVAL LOCK

- If a route must intersect flood water (PLAN-002), autonomous dispatch is locked.
- System forces a high-priority amber Commander Review Gate in the UI.
- Zero-hallucination guarantee: AI cannot spoof GPS, fabricate stock, or bypass locks.

ENTERPRISE ARCHITECTURE

Battle-Tested, Enterprise-Grade Modern Stack

Next.js 16, Custom MapLibre Spatial Engine, FastAPI Microservices, and LangGraph Orchestration

TACTICAL FRONTEND

- Next.js 16 (App Router) & React 19 optimized for 24/7 command centers.
- Custom MapLibre GL JS spatial engine on CartoDB Dark Matter tiles.
- Live GeoJSON hazard rendering, pulsing disaster pins & smooth fitBounds zooming.

ASYNC CORE & LANGGRAPH

- High-throughput FastAPI microservices in Python 3.12 with async workers.
- LangGraph StateGraph governing cyclical operational agent state machine:
- extract -> validate -> monitor -> assess -> plan -> await_approval -> dispatch.

PERSISTENCE & FIELD EDGE

- PostgreSQL (Neon Serverless) with ACID transactions & idempotent locks.
- Telegram Bot API gateway delivering turn-by-turn routes directly to driver phones.
- Real-time driver cryptographic delivery acknowledgement verification.

MISSION CONTROL DEMO

Live Tactical Command Experience: Bangalore South Pilot

Real-Time Spatial Intelligence, AI Extraction Drawers, and One-Click Mission Approval

TACTICAL #0a0f1d COCKPIT

- High-contrast, dark-mode interface built for zero cognitive fatigue.
- Real-time KPI telemetry bar: Active Floods, Pending Gate, Dispatched Units.
- Programmatic MapLibre camera transitions auto-centering on emergency hot-spots.

5 LIVE PILOT SCENARIOS

- PLAN-001 (Dispatched): 850 relief units tracked in real time to Chandapura hub.
- PLAN-002 (Safety Lock): 2.1m flood intersection triggers Commander Approval Lock.
- PLAN-003 (Detour): Critical insulin route dynamically arcs 2.2km around Marsur basin.

SUB-300MS DISPATCH PUSH

- Side-by-side Gen-AI drawer: unedited citizen voice SOS vs. structured 98% JSON.
- One-click commander approval unlocks safety gate and dispatches field driver.
- Telegram driver bot pushes turn-by-turn navigation coordinates instantly.

NATIONAL VISION & SCALE

The National Disaster Nervous System: Pan-India NDMA Rollout

Scaling from Regional Flood Corridors to Unified National Multi-Agency Disaster Command

DRONE SWARMS & RADAR

- Direct ingestion pipeline for ISRO RISAT & Sentinel-1 radar imagery.
- Autonomous flood mapping through torrential rain and thick monsoon cloud cover.
- Aerial UAV drone swarm feeds providing live road passability verification.

PAN-INDIA MULTI-TENANCY

- Multi-tenant architecture scaling across all 36 States and Union Territories.
- Pre-built integration connectors for Agmarknet grain silos and Red Cross depots.
- Granular role-based access for NDRF, SDRF, and District Magistrates.

MULTIMODAL AUTONOMY

- Expands beyond 4x4 trucks to NDRF motorized boats and amphibious ARVs.
- Automated heavy-payload cargo drone dispatch for isolated rooftop rescues.
- ReliefRoute transforms disaster management into an autonomous life-saving machine.